Introduction to High-Energy Astrophysical objects

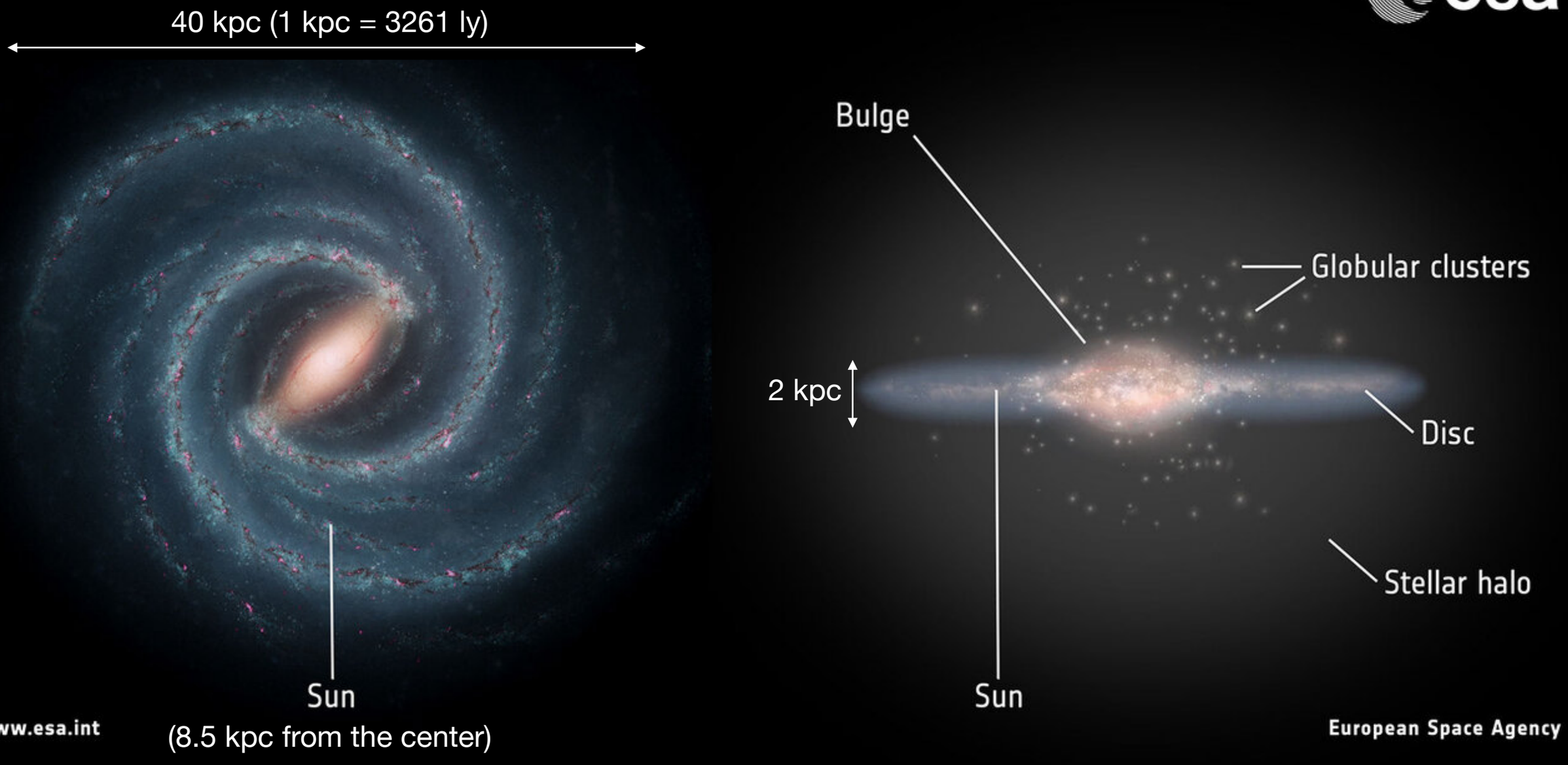
(Lecture 1)

Satyendra Thoudam
Khalifa University

Outline of this lecture

- **Anatomy of the Milky Way galaxy**
- **Interstellar/intergalactic medium**
- **Galactic objects:**
  - Formation and evolution of stars
  - Supernova
  - Neutron stars and Pulsars
  - Black holes
  - Binaries
- **Extragalactic objects:**
  - Active Galactic Nucleus
  - Gamma-Ray Burst

Milky Way Galaxy



www.esa.int

European Space Agency

Interstellar medium

*** Interstellar medium has the following main components:**

- Matter (atoms, molecules, ions)
- Photons (microwave, infrared, optical etc.)
- High-energy particles (all nuclei, electrons)
- Magnetic field (regular and random components)

*** All have almost the same energy density ~ 1 eV/cm^3 (equipartition?).**

*** Indicating some sort of interactions happening between the different components.**

Interstellar medium

The interstellar matter consists of gas and dust.

Gas is atoms and small molecules, mostly hydrogen and helium. (90%H, 9%He, <1% rest of the elements)

Dust is more like soot or smoke; larger clumps of particles. (Less abundant in the interstellar space)

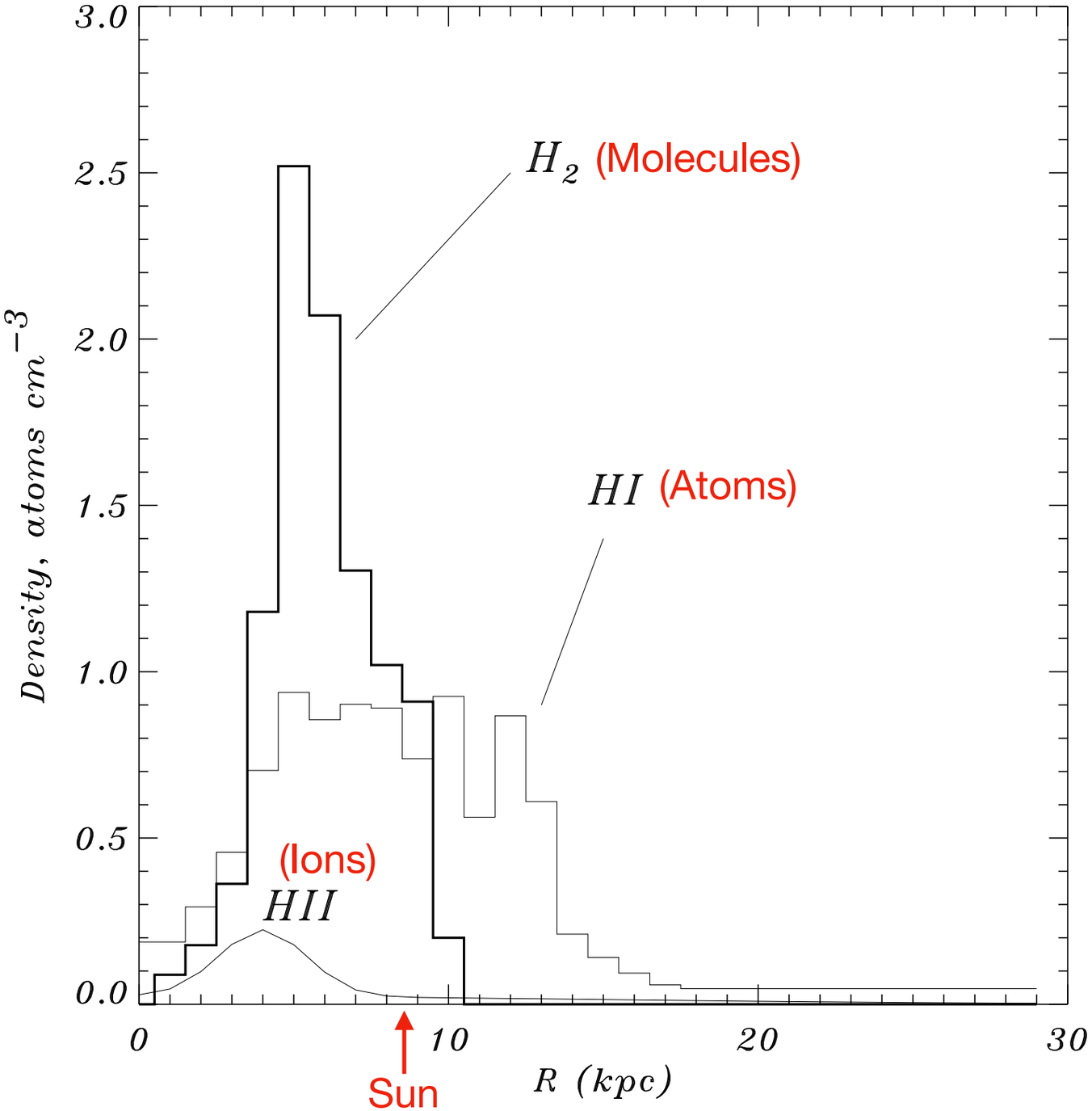


A wide-angle view of the Milky Way—the dark regions are dust clouds, blocking light from the stars beyond

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Interstellar medium

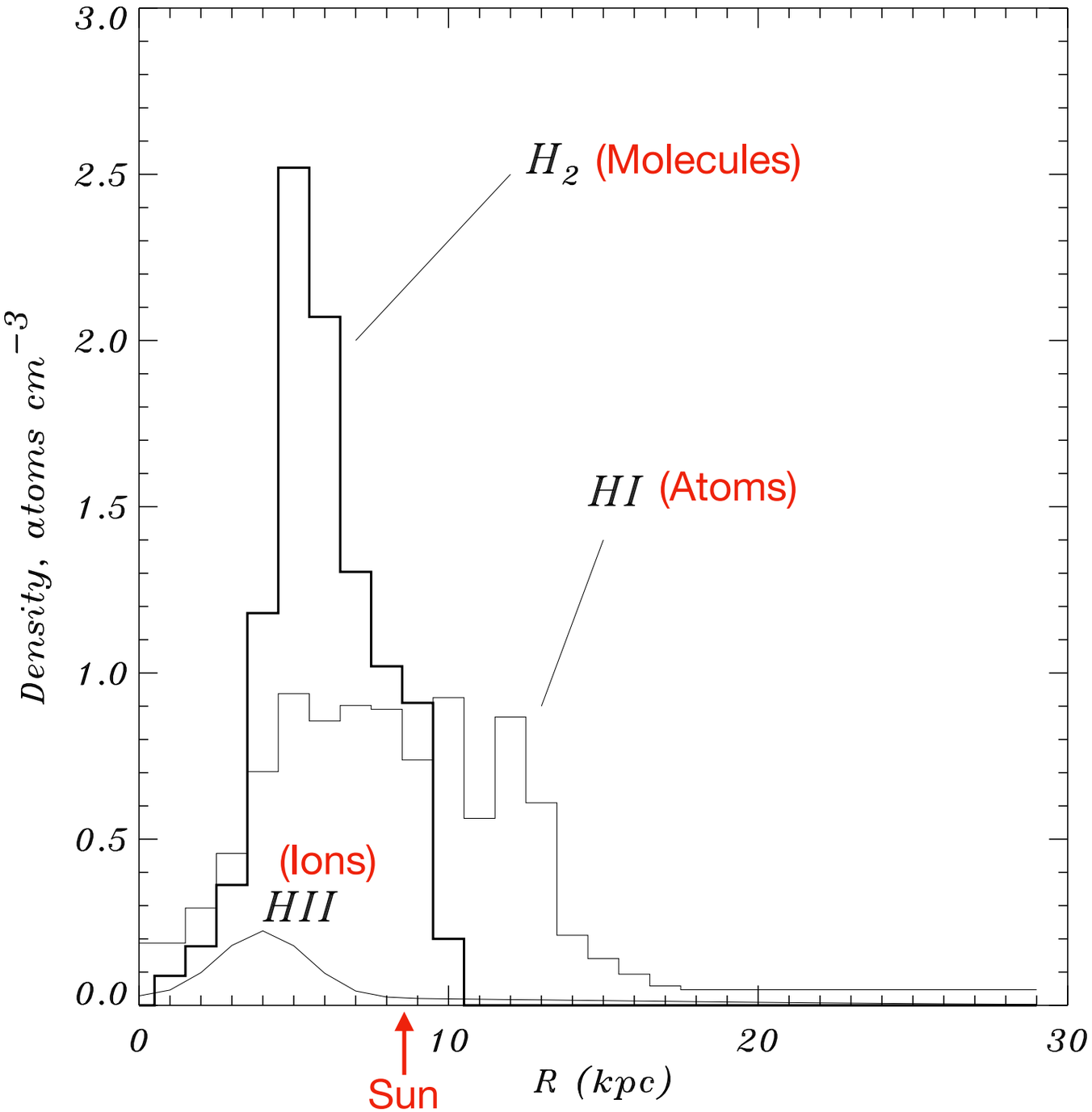
Radial distribution of matter at the Galactic disk (~ 1 atom/cm³)



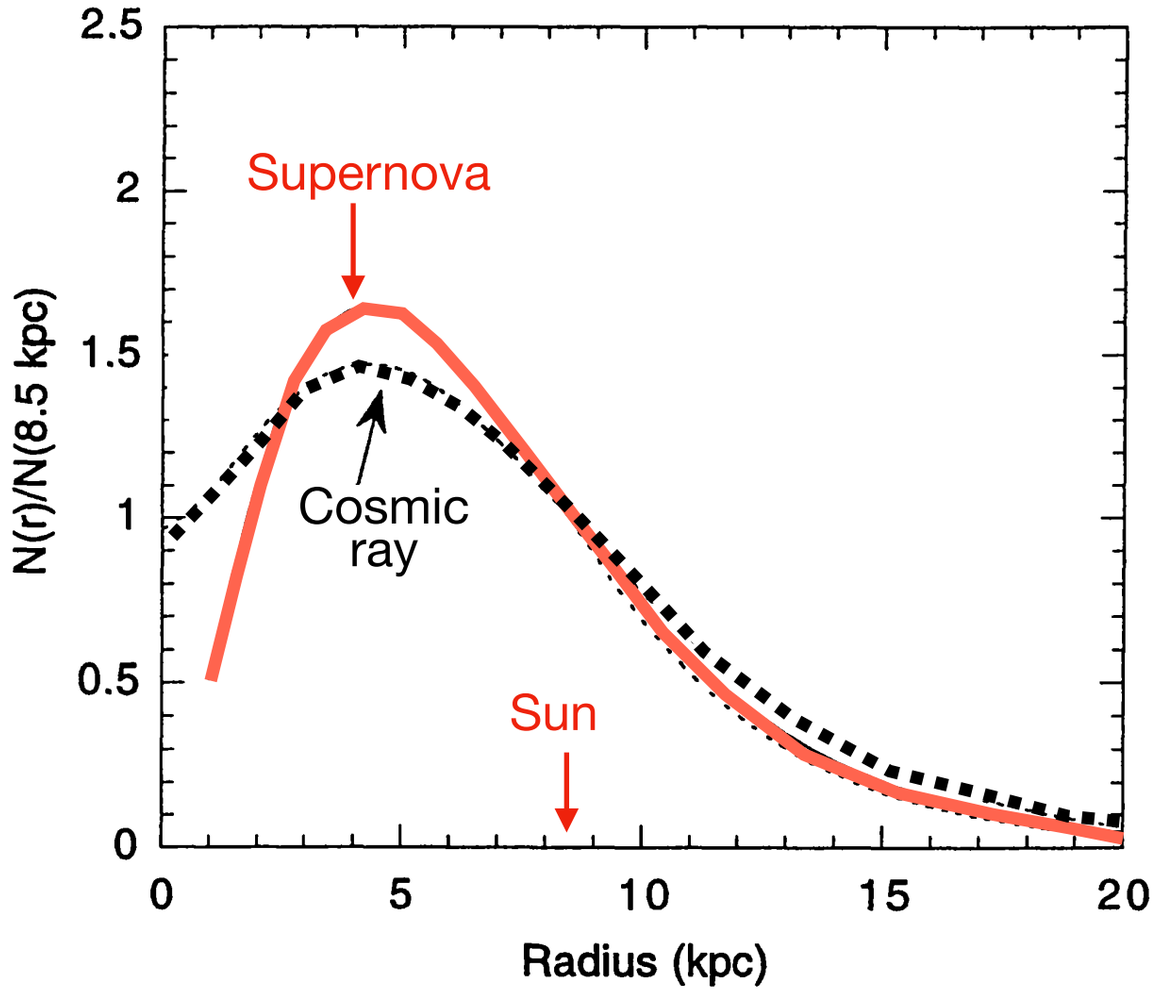
Strong & Moskalenko, 1998, ApJ, 509, 212

Interstellar medium

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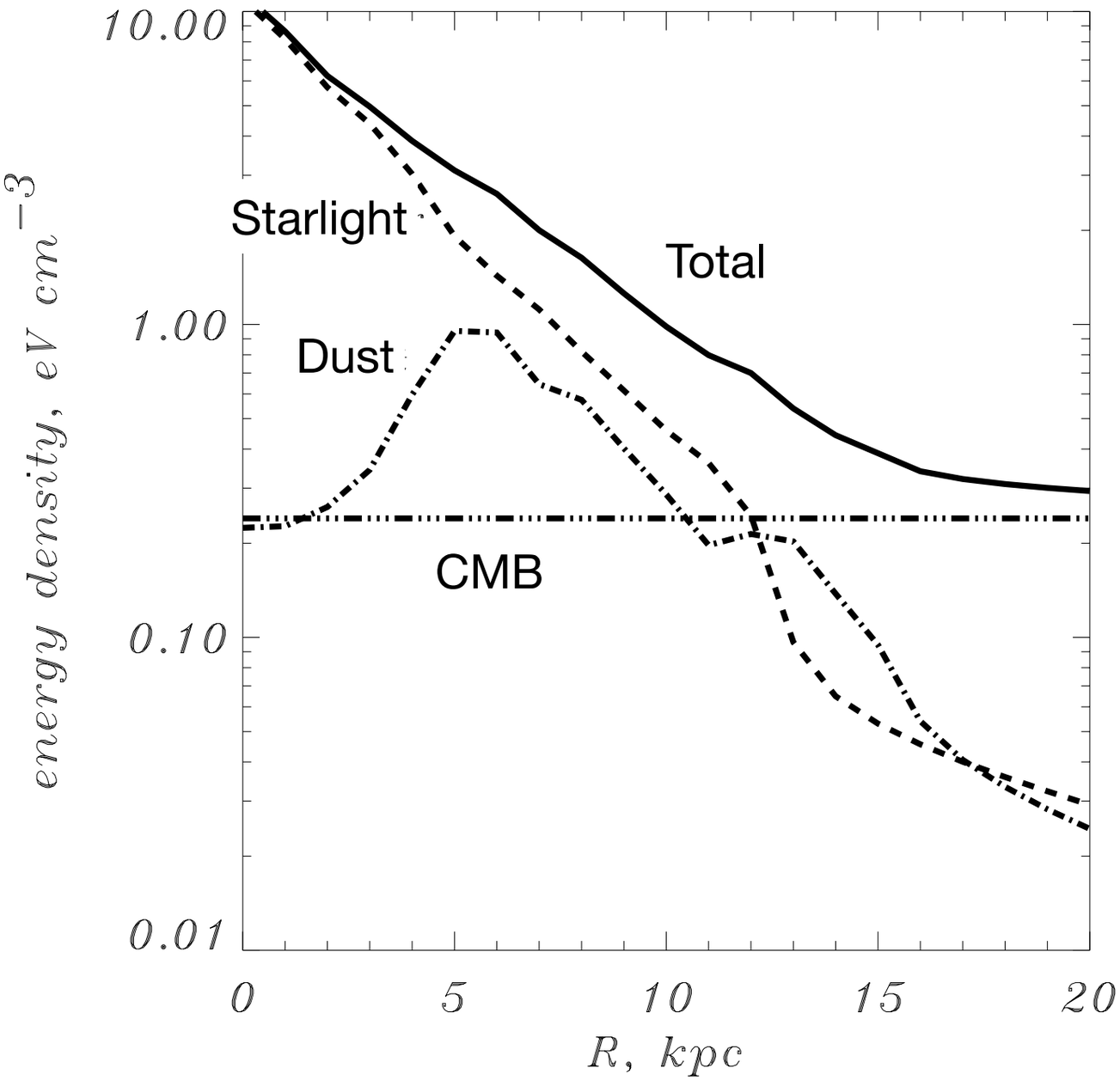
Radial distribution of supernova remnants (stars) and cosmic rays at the Galactic disk



Case & Bhattacharya, 1996, A&A, 120, 437

Strong & Moskalenko, 1998, ApJ, 509, 212

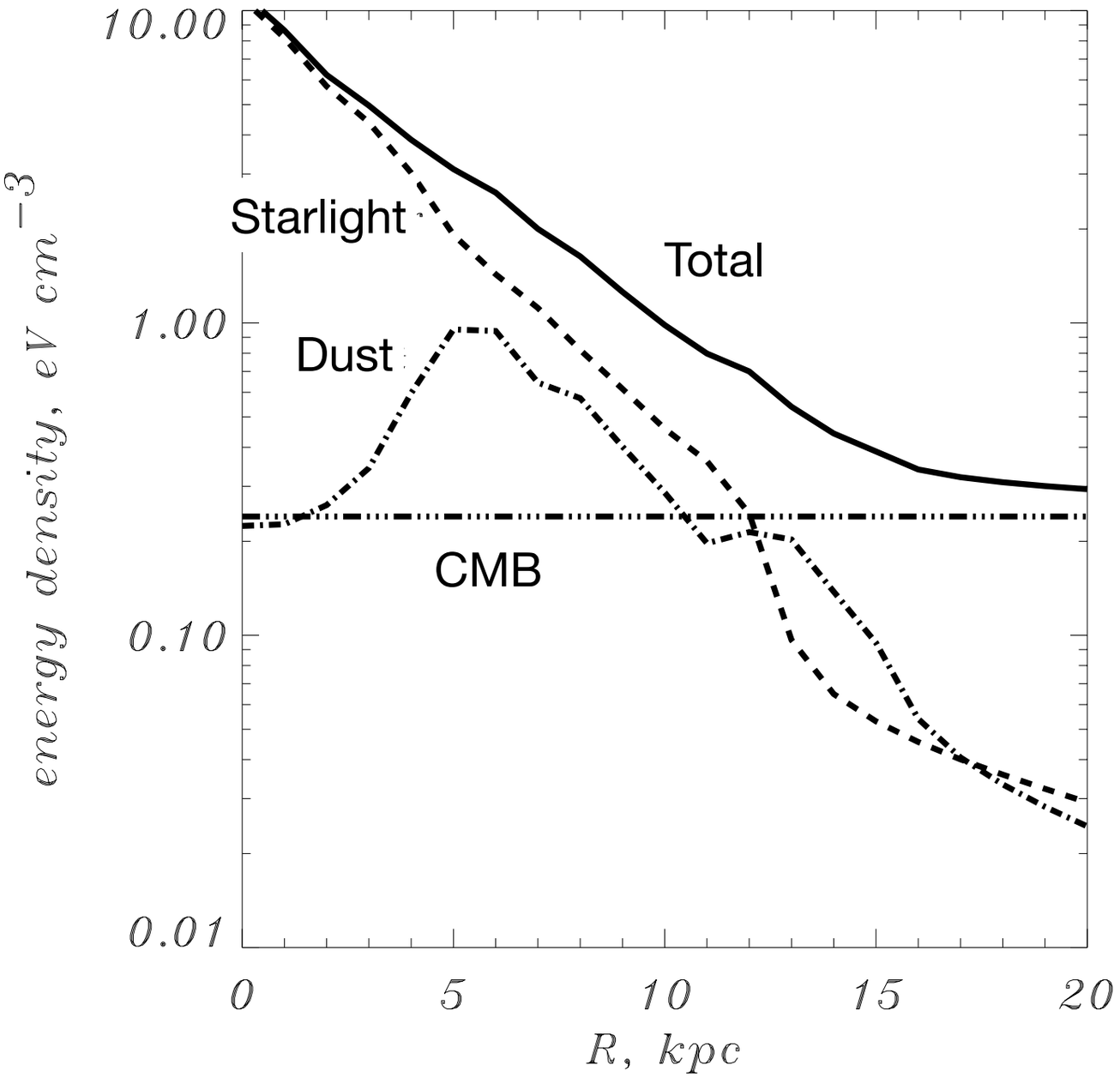
Interstellar radiation energy density as a function of radial distance at the Galactic disk



Strong+ , 2000, ApJ, 537, 763

Interstellar medium

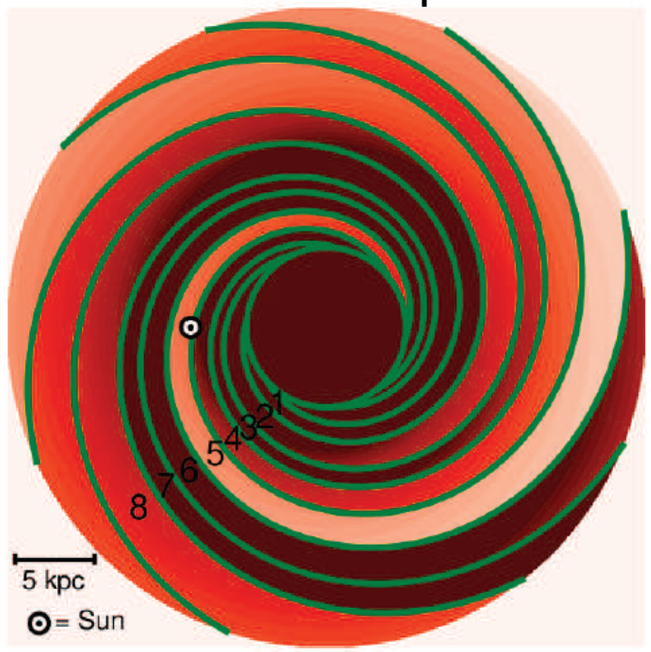
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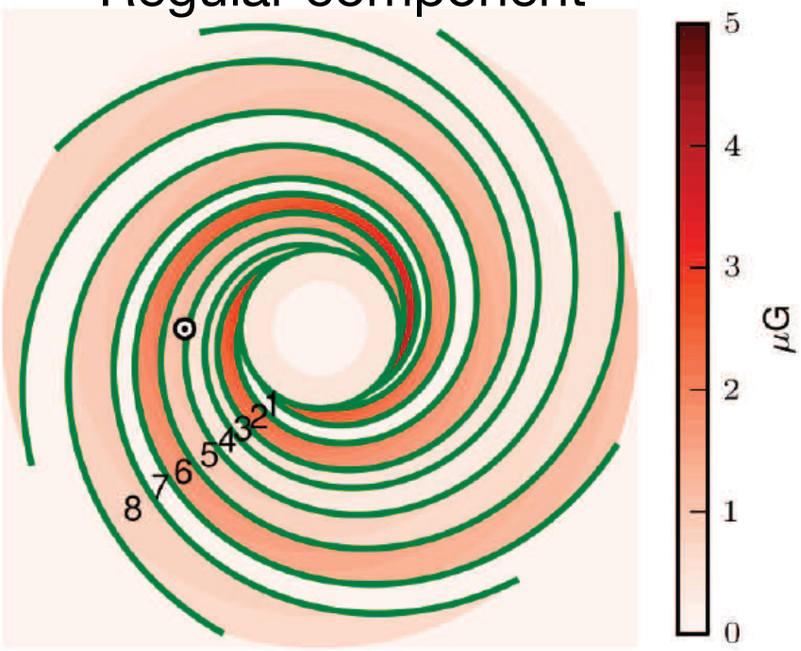
Strong+ , 2000, ApJ, 537, 763

Magnetic field at the Galactic disk (B~ 1-5 μG ~ 10^{-6} G)

Random component



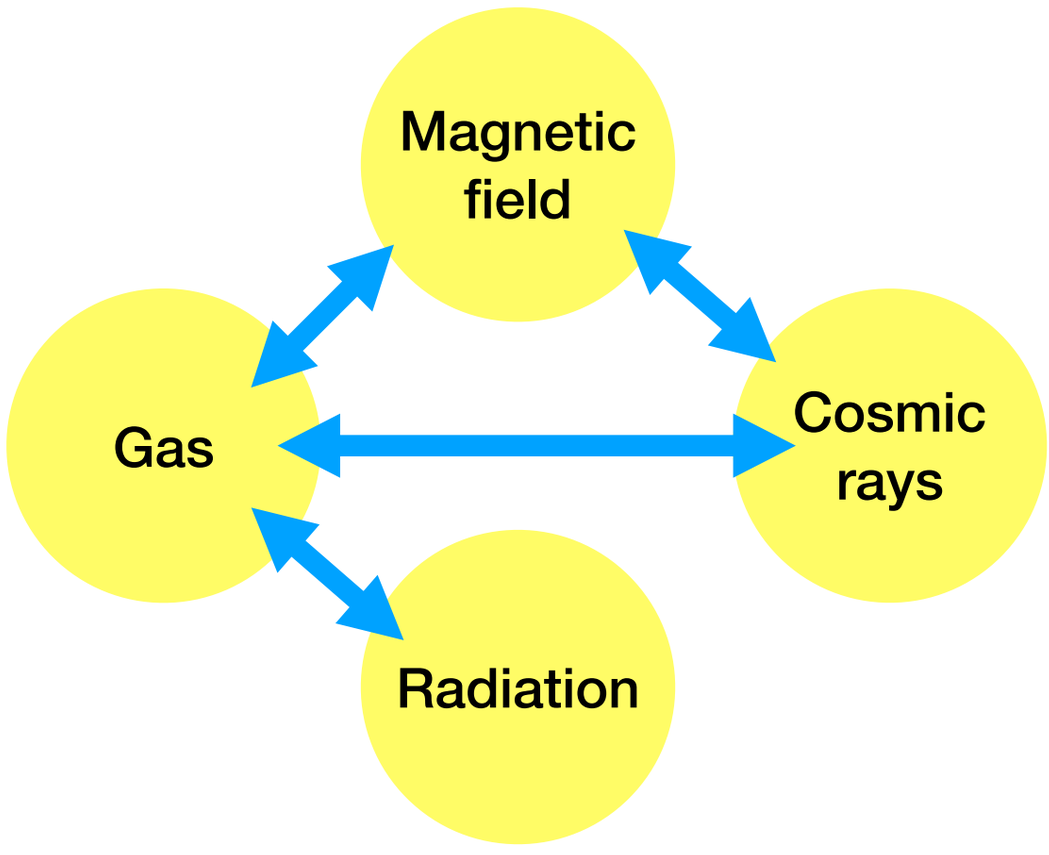
Regular component



G. Farrar, Comptes Rendus Physique 15 (2014) 339-348

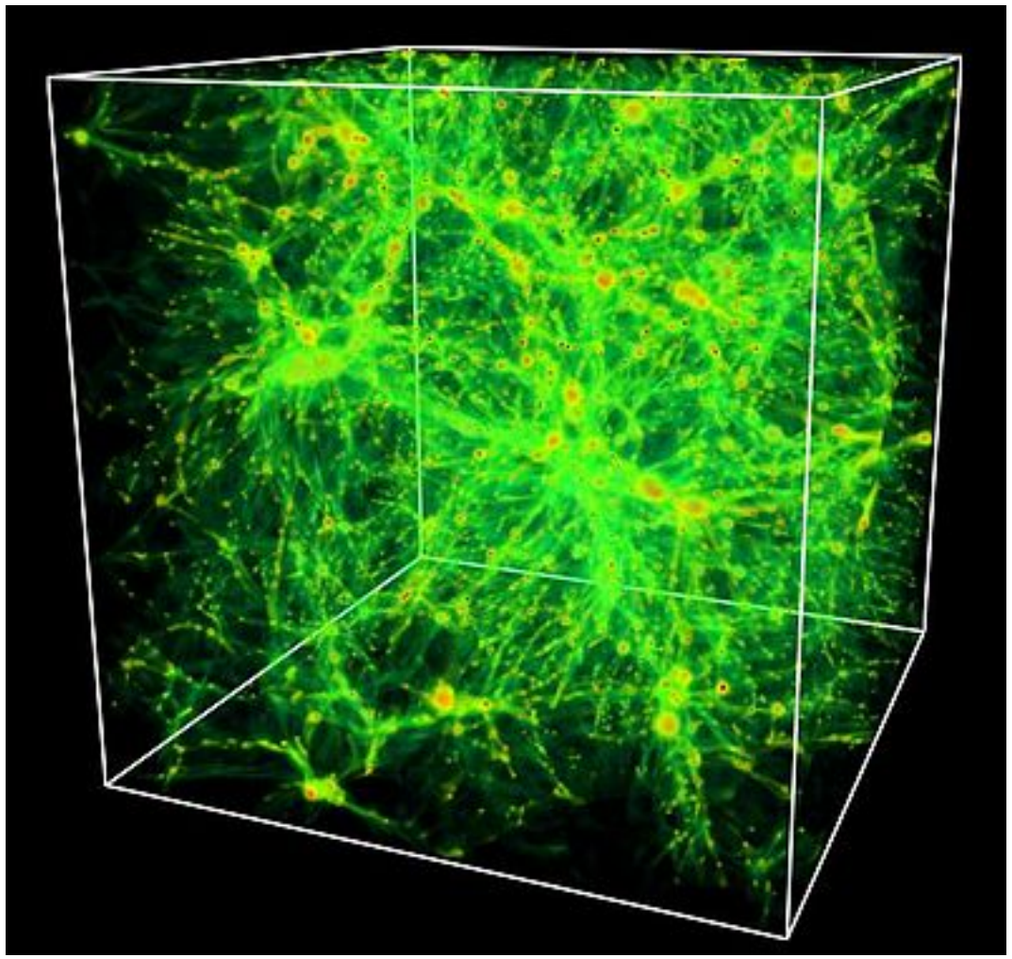
Equipartition in energy

- Energy density: $W_{CR} \sim W_{mag} \sim W_{gas} \sim W_{rad} \sim 1$ eV/cm^3.
- This indicates interaction between the different components of the ISM.



Intergalactic medium

- Space between galaxies.
- Not much is known about the extragalactic medium.
- Gas density ~ 1 atom/m^3 , mainly ionized hydrogen.
- Magnetic field ~ nanoGauss (nG) = 10^{-9} G.

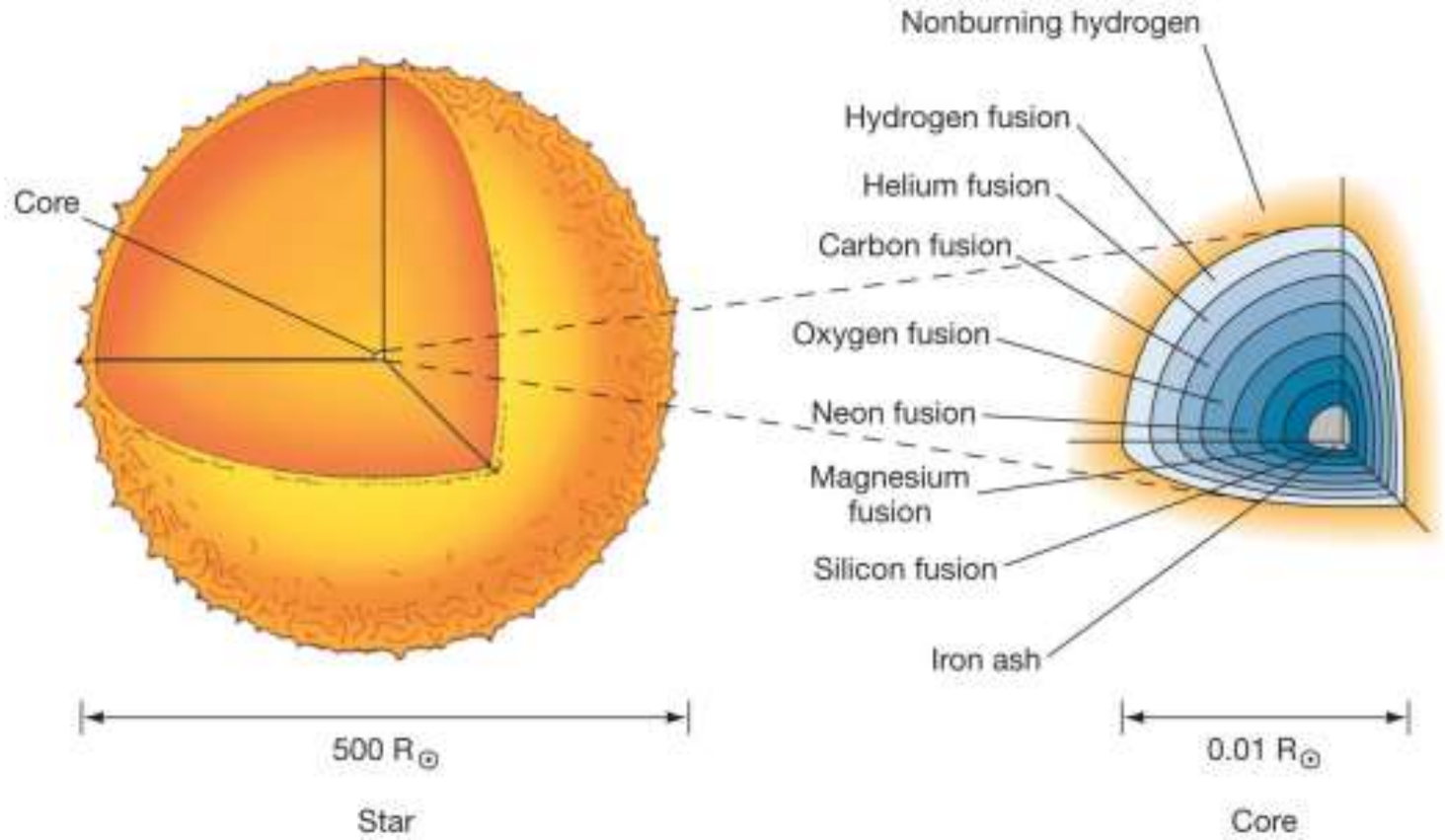


Credit: Wikipedia

Computer simulation showing the distribution of warm-hot intergalactic gas.

Galactic high-energy astrophysical objects

Formation and evolution of high-mass stars (≳10 Solar mass)



- A high-mass star can fuse element in its core up to the formation of iron nuclei.
- Iron is the most stable element. No fusion possible beyond iron.
- Not sufficient outward pressure -> Rapid gravitational collapse.

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Galactic high-energy astrophysical objects

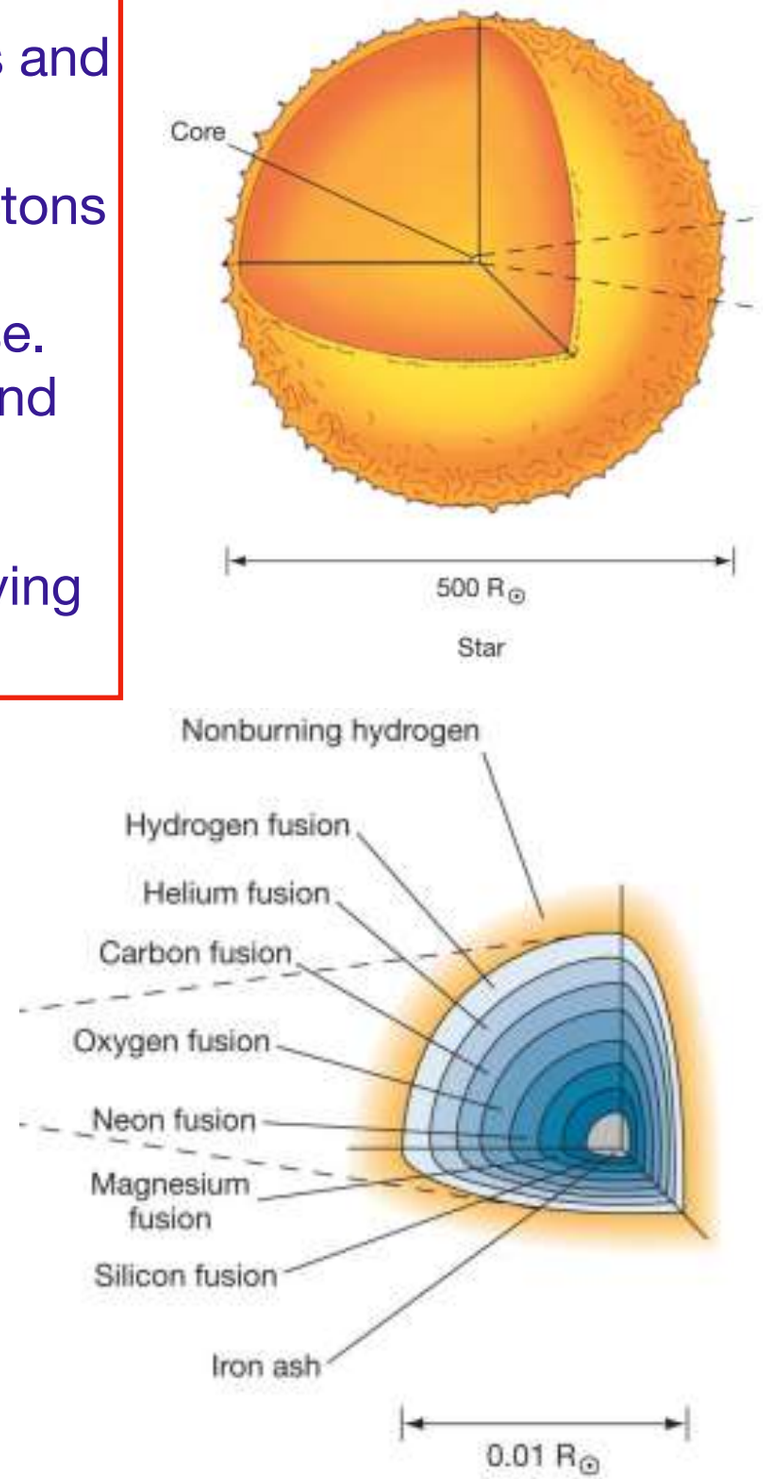
Supernova (Type II or core-collapse): The end of a high-mass star

- Core temperature rises to about 10 billion K.
- Photons can split iron into lighter nuclei and eventually into protons and neutrons, cooling the core (Photo-disintegration).

- Ultimately, core contains only protons, neutrons, electrons and photons in very high densities.
- As the core continues to contract, it becomes more and more dense.
- The protons and electrons are crushed together to form neutrons and neutrinos (Neutronization):

$$p + e \longrightarrow n + \text{neutrino}$$

- Neutrinos escape the core as they hardly interact with matter, carrying away energy, further reducing the internal pressure support.



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Galactic high-energy astrophysical objects

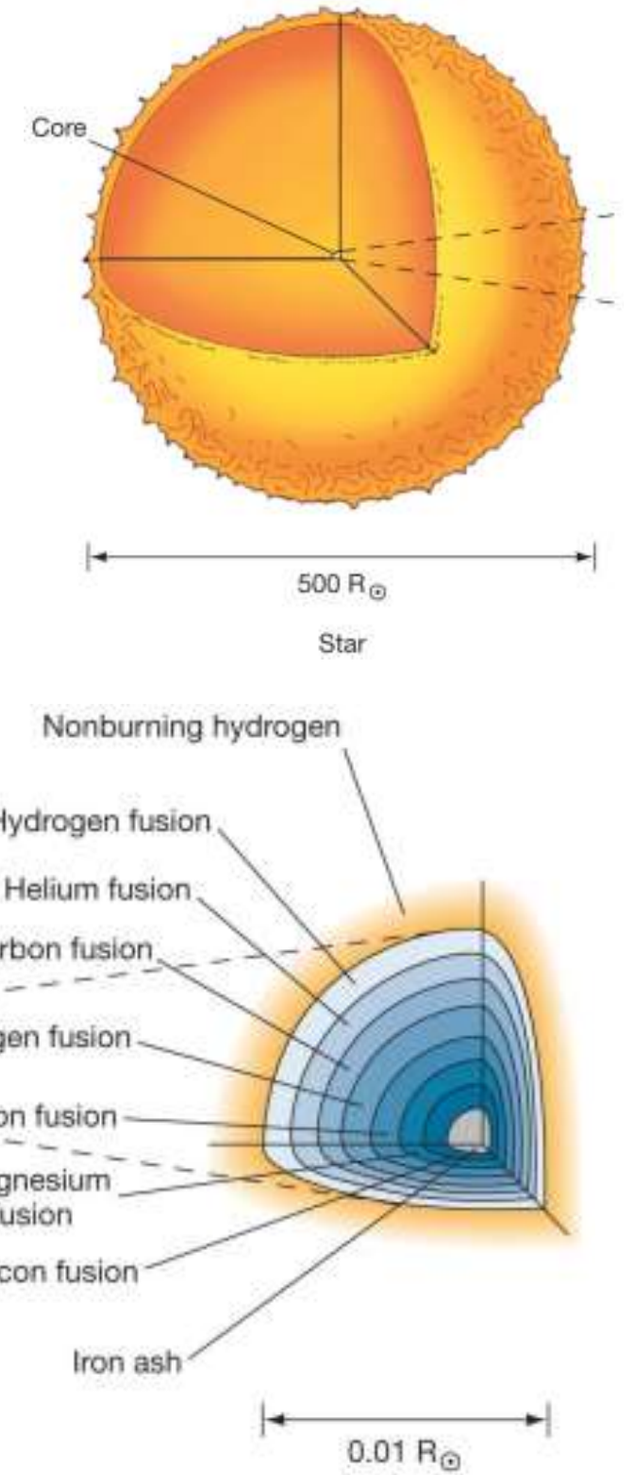
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The neutrinos escape; the neutrons are compressed until they come into contact with one another. At this stage, the core has a density of about 10^15 kg/m3.

The neutrons starts resisting the collapse, and finally completely stops the contraction when the density is as high as ~10^(17-18) kg/m3. Then, the core rebounds, creating a (core-collapse) supernova explosion.

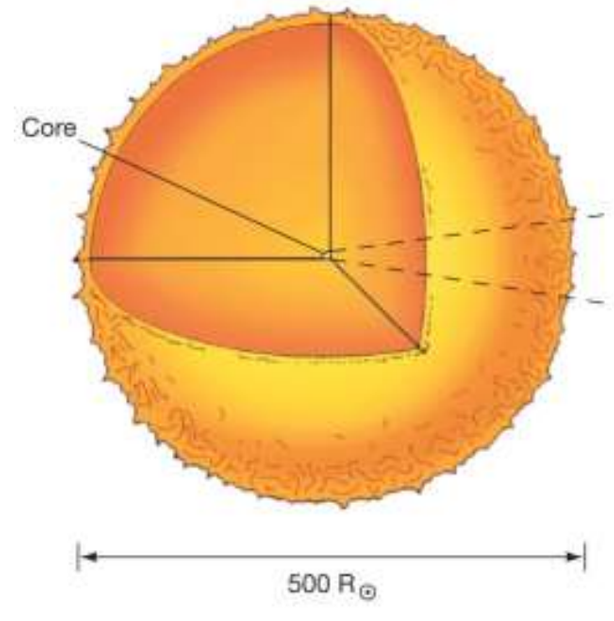
This final stage of collapse and rebound takes only about a second!

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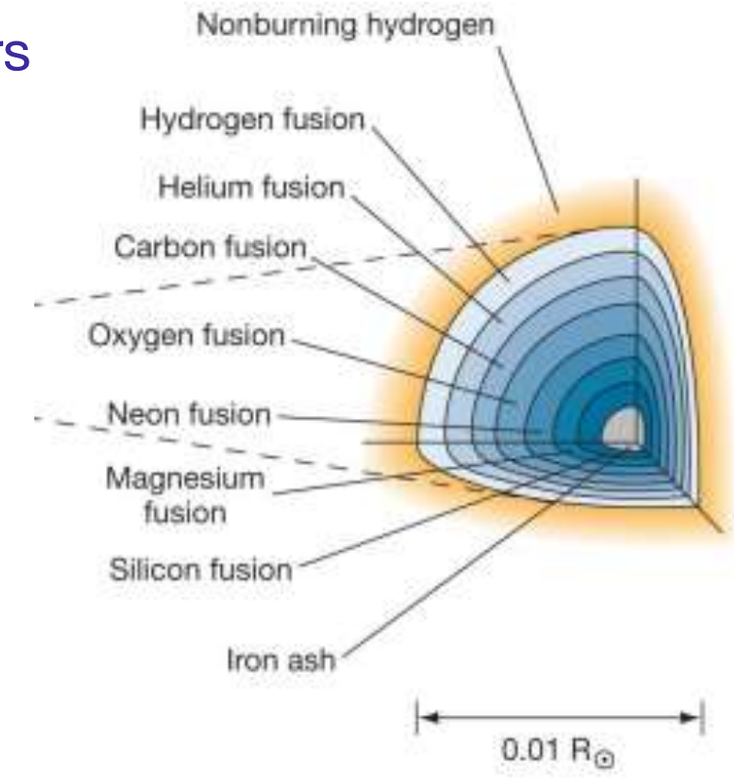
Galactic high-energy astrophysical objects

Supernova (Type II or core-collapse): The end of a high-mass star

- A strong shock wave sweeps through the star at very high speed (~ 10^4 km/s).
- Blast all the overlying layers of heavy elements formed by fusion process.
- Supernova explosion releases an enormous amount of energy (~10^{53} ergs).
- The exploding star can be easily distinguish from the rest of the stars in the galaxy due to its brightness.

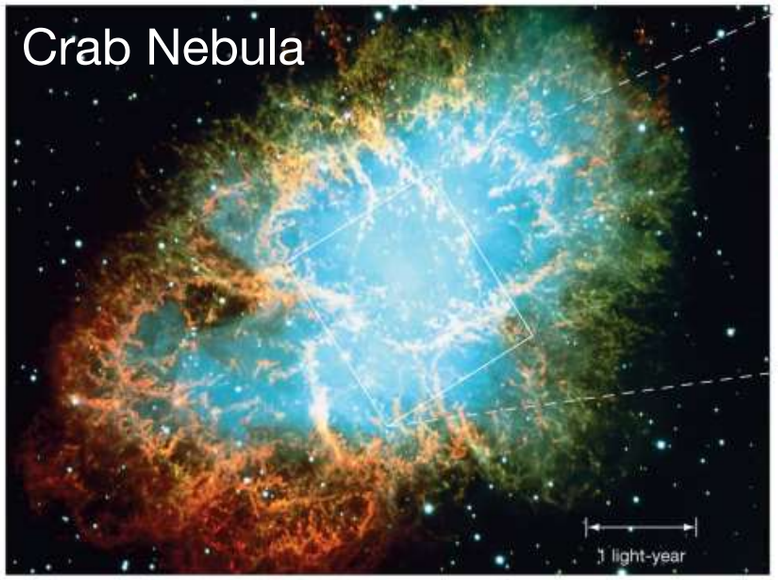


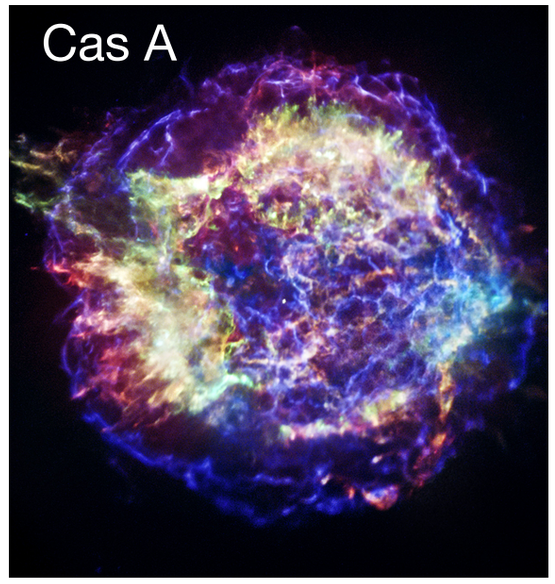
- Supernova explosions happen at the rate of ~ 1 in 30 to 100 years in our Galaxy.

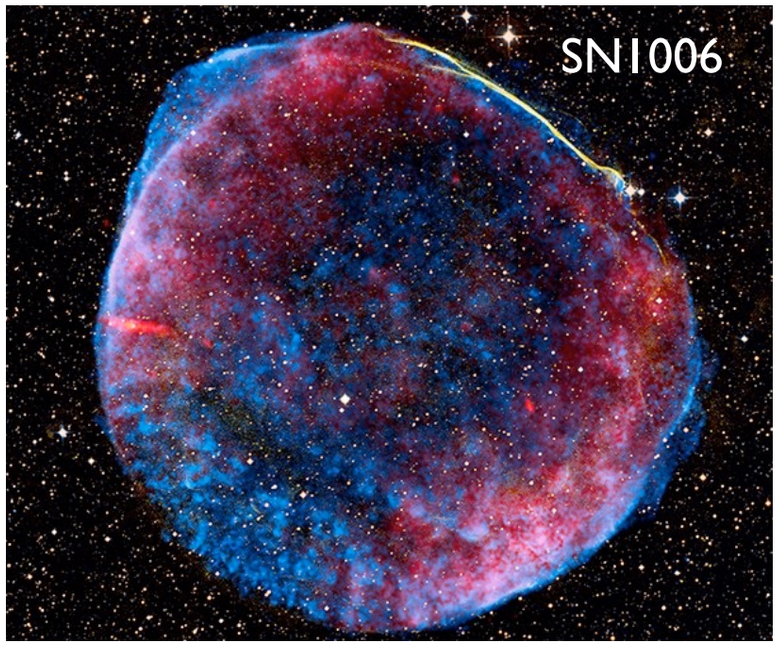


Galactic high-energy astrophysical objects

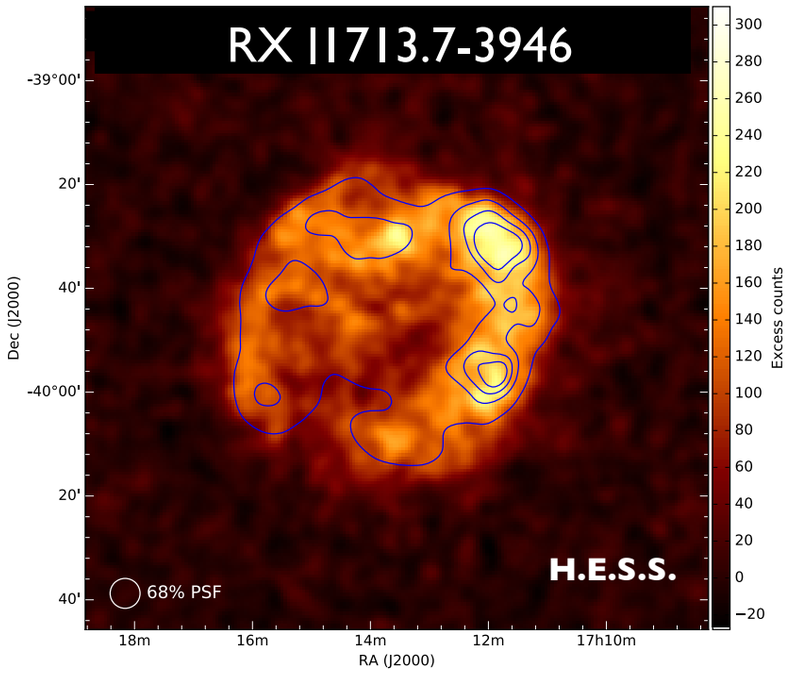
Supernova remnants (SNRs): Remains of Type II supernova explosions







Credit: NASA/ESA/HESS



- SNRs are considered as the most potential source of high-energy cosmic rays.

Neutron stars and pulsars: Remains of Type II supernova explosions

After a Type II supernova, part of the core may survive. It is very dense—as dense as an atomic nucleus—and is called a neutron star.

Neutron stars, although they have 1–3 solar masses, are so dense that they are very small. This image shows a 1-solar-mass neutron star, about 10 km in diameter, compared to Manhattan:



Density: 10^17 -10^18 kg/m3

Density of an atomic nucleus: 3x10^17 kg/m3

Neutron degeneracy pressure supports the neutron star.

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Neutron stars and pulsars: Remains of Type II supernova explosions

Properties of a neutron star:

- Rotation—as the parent star collapses, the neutron core spins very rapidly, conserving angular momentum. Typical periods are fractions of a second.

- Magnetic field—again as a result of the collapse, the neutron star's magnetic field becomes enormously strong. Field lines get squeezed as the core collapses.

Galactic high-energy astrophysical objects

Neutron stars and pulsars: Remains of Type II supernova explosions

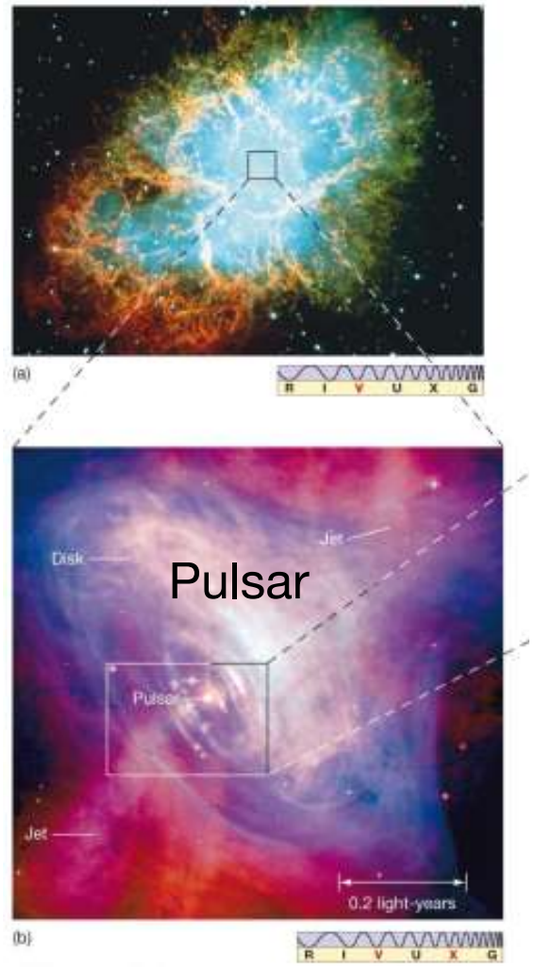
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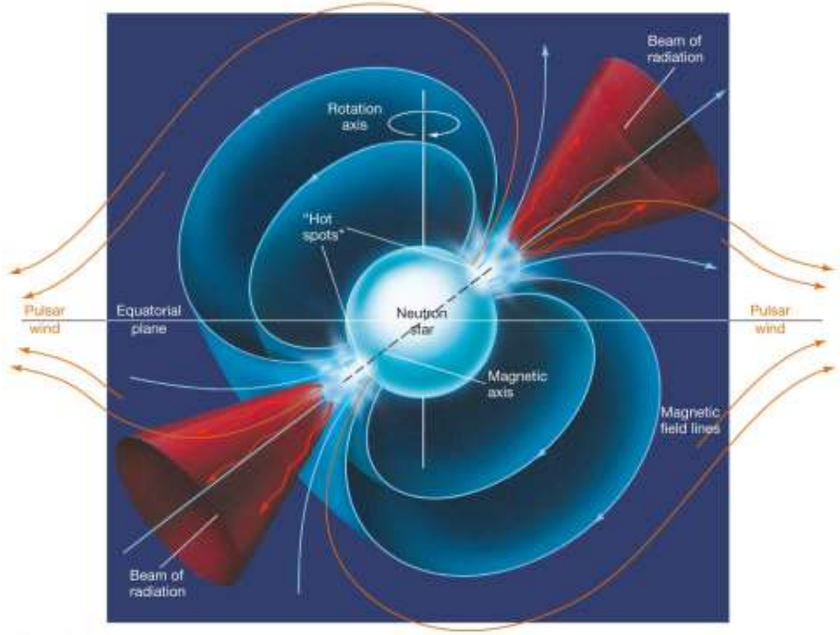
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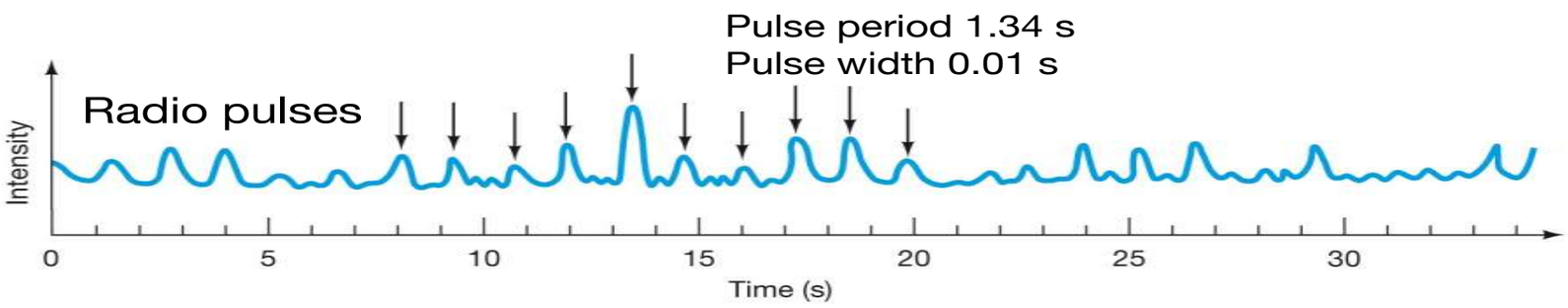
Crab Nebula



Pulsars: Pulse emission (lighthouse effect)

Strong jets of matter are emitted at the magnetic poles. If the rotation axis is not the same as the magnetic axis, the two beams will sweep out circular paths. If the Earth lies in one of those paths, we will see the star pulse.





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Galactic high-energy astrophysical objects

Black Holes: Remains of a star having > 25 Solar mass

The mass of a neutron star cannot exceed about 3 solar masses. If a core remnant is more massive than that, nothing will stop its collapse, and it will become smaller and smaller and denser and denser.

Eventually, the gravitational force is so intense that even light cannot escape. The remnant has become a black hole.

- This happens mostly when the mass of a star exceeds 25 Solar mass.
- The 3 Solar mass limit can be even higher in reality if magnetic field and rotation are taken into account.

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Galactic high-energy astrophysical objects

Black Holes: Remains of a star having > 25 Solar mass

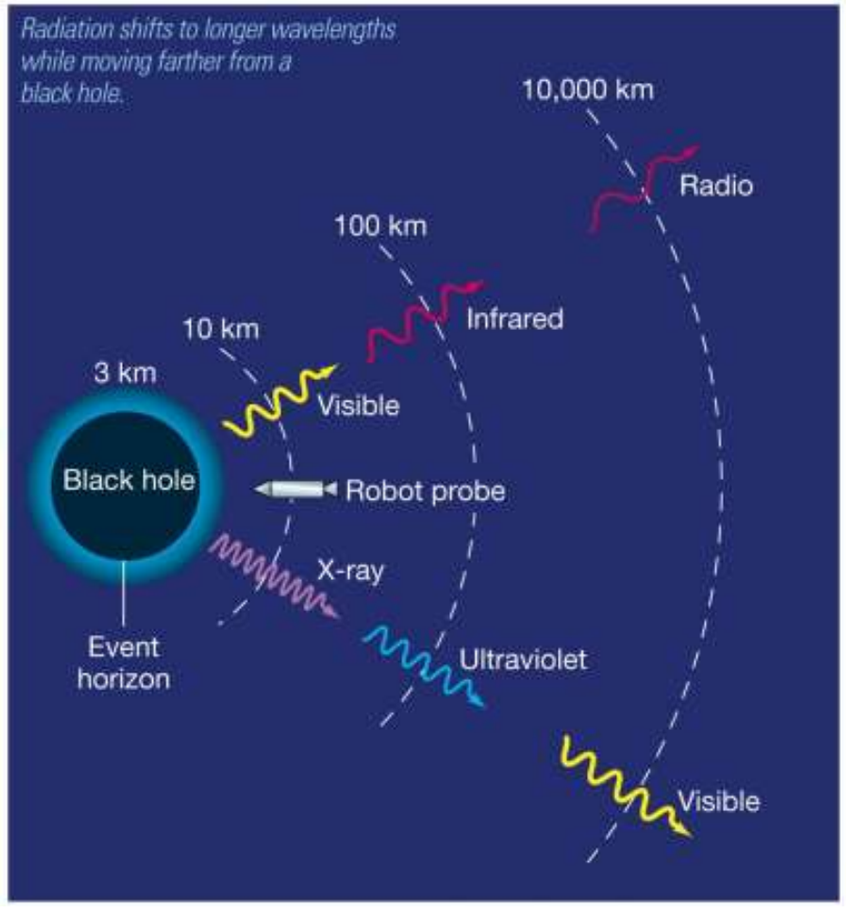
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Gravitational redshift

A photon escaping from the vicinity of a black hole has to lose energy:
=> Decrease in frequency
=> Increase in wavelength



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First image (shadow) of black hole

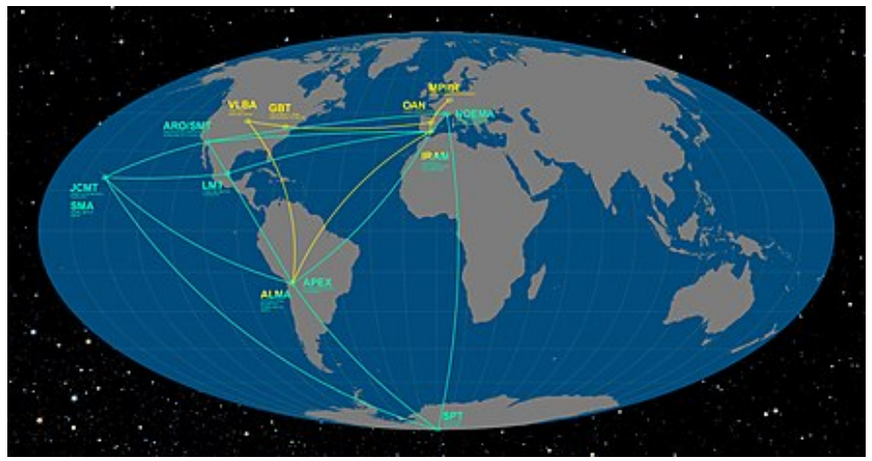
Event Horizon Telescope:

- Array of global network of radio telescopes
- Wavelength ~ 1.3 mm
- Angular resolution ~ 25 micro arc seconds
- Targets: Blackholes in Milky way & M87

Messier 87

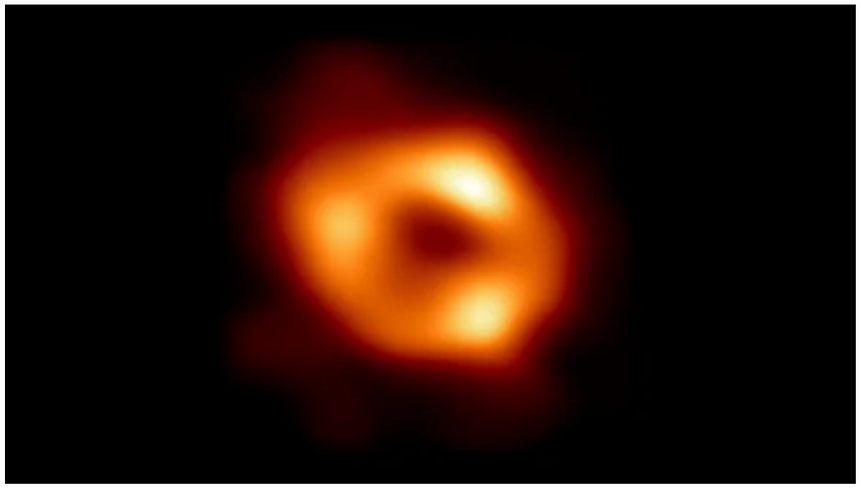


April10
2019



Event Horizon Telescope
https://eventhorizontelescope.org

Milky Way

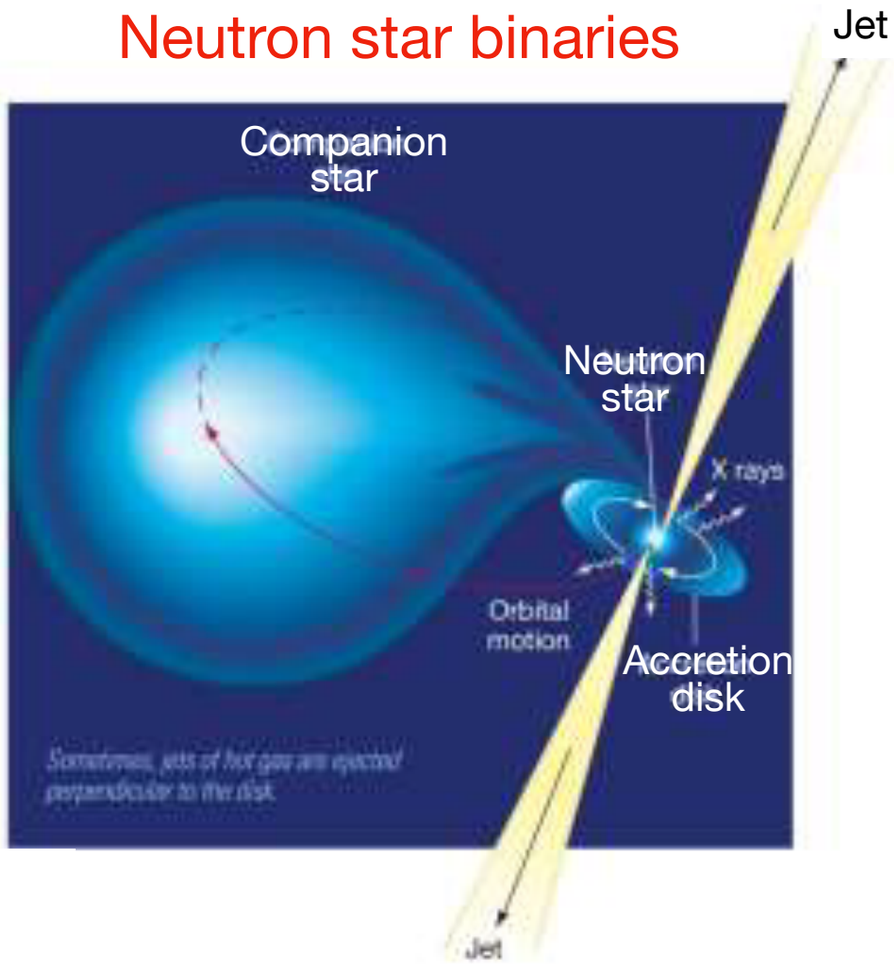


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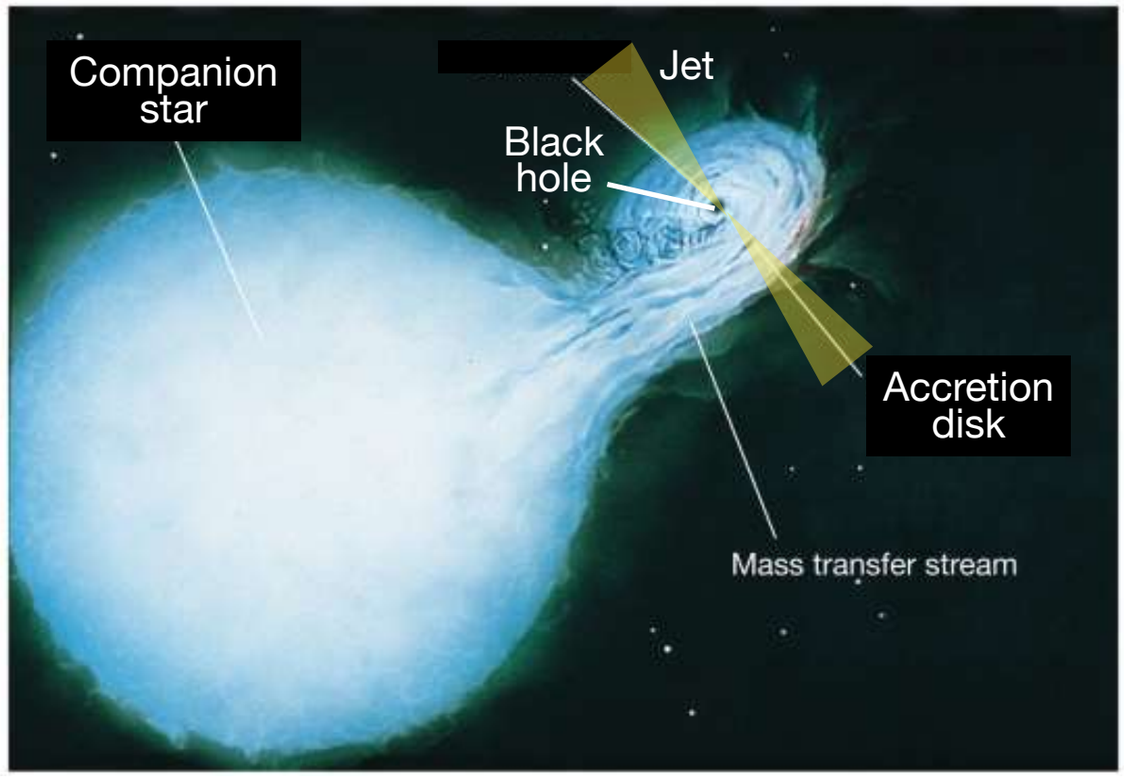
Galactic high-energy astrophysical objects

Binary systems

Neutron star binaries

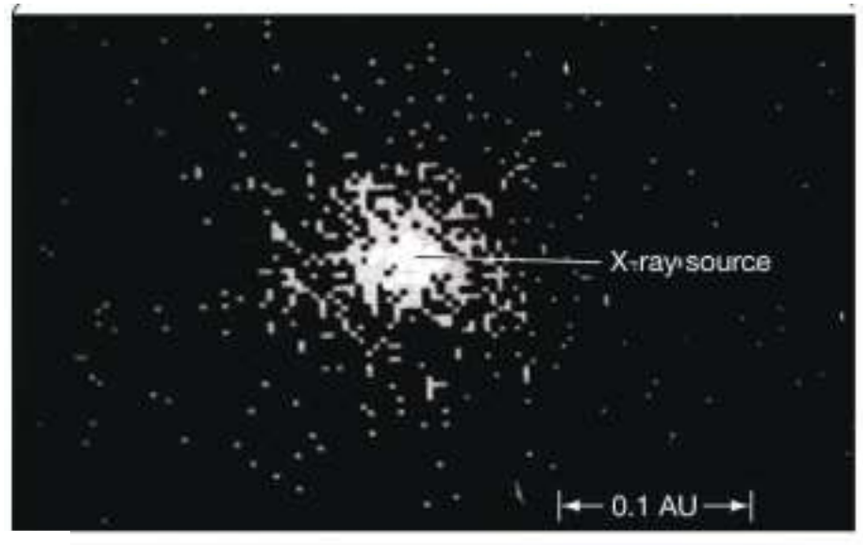


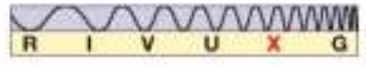
Black hole binaries



Cygnus X-1 is a very strong black-hole candidate:

- Its companion is about 25 solar masses.
- The black hole candidate has a mass of about 10 solar mass.
- Hot gas appears to flow from the visible companion to the unseen black hole, emitting strong X-ray signals.





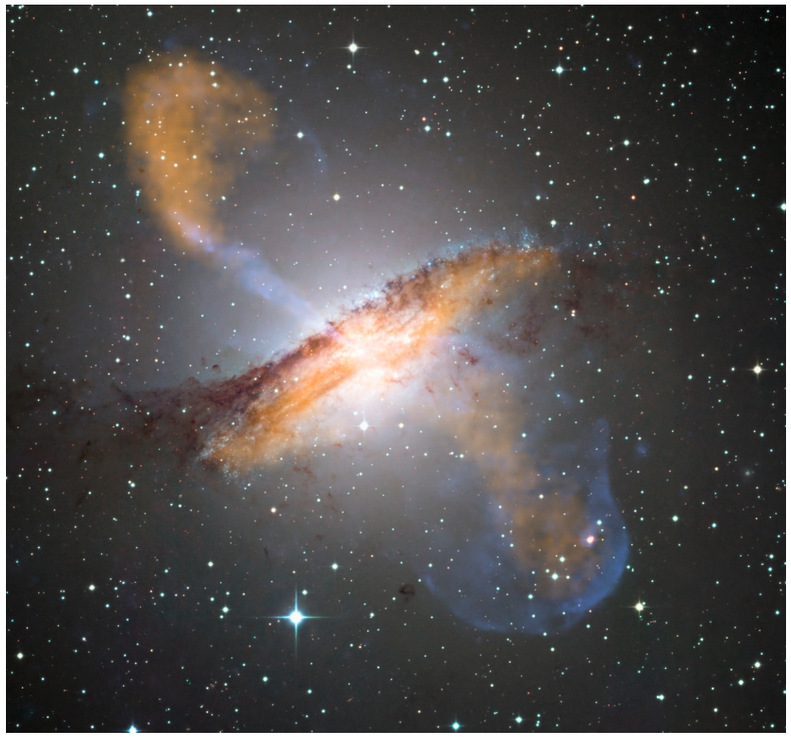
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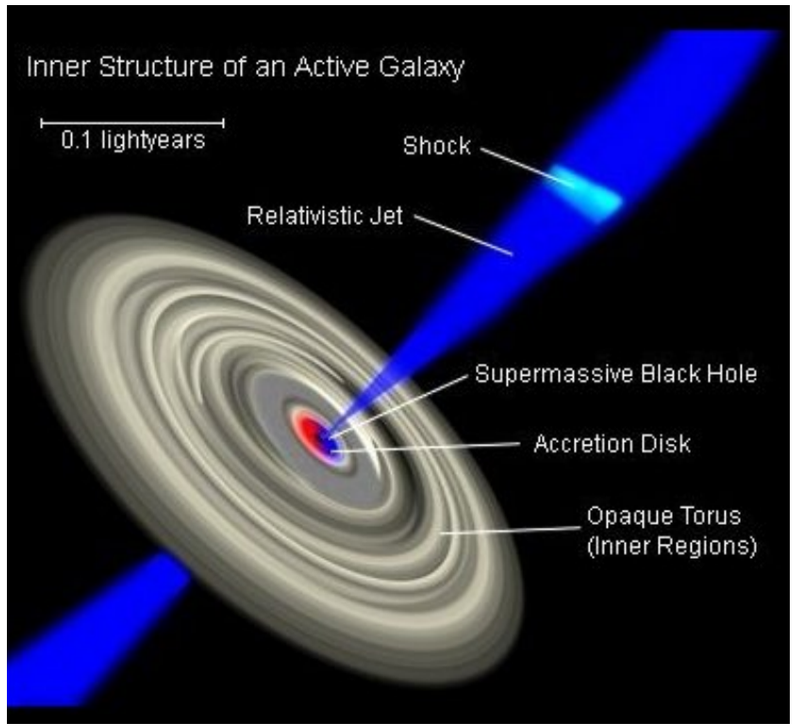
Extragalactic objects

Active Galactic Nucleus (AGN)

- The centers of galaxies like ours contain black holes of masses about 1 million solar mass.
- Some galaxies host black holes of about 100 million solar masses. The compact central region of such galaxies are called Active galactic nucleus (AGN).
- Typical size of an AGN ~ size of our Solar System.
- An AGN typically releases a luminosity of about 10^{45} erg/s which is about 100 times larger than the luminosity of our galaxy.
- AGNs are a candidate source of ultra high-energy cosmic rays.
- Centaurus A is one of the closest AGNs to the Earth.
- Blazars: AGNs whose jet points towards the Earth.

Credit: ESO/WFI



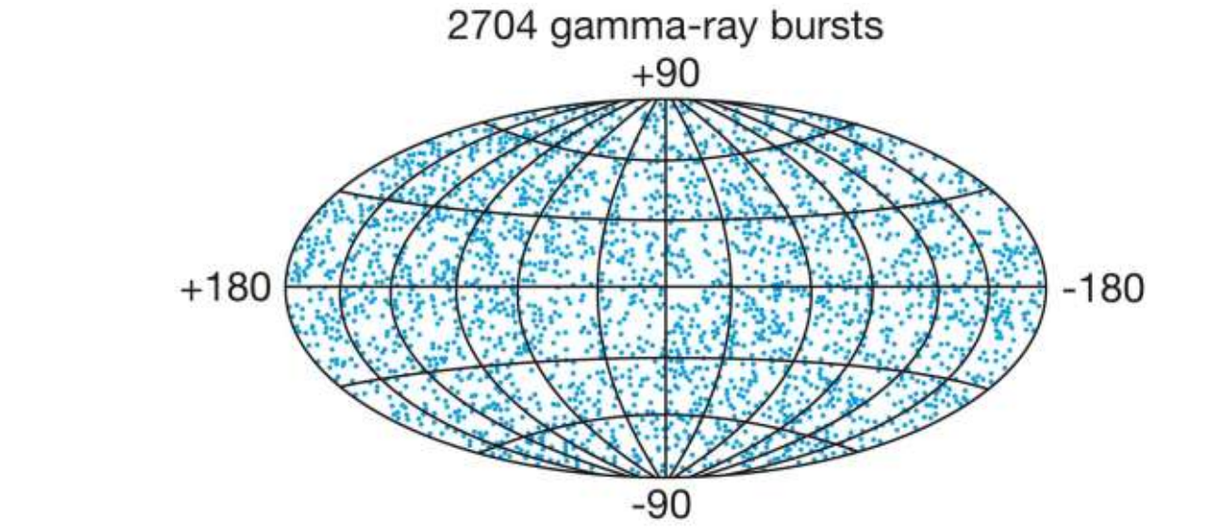


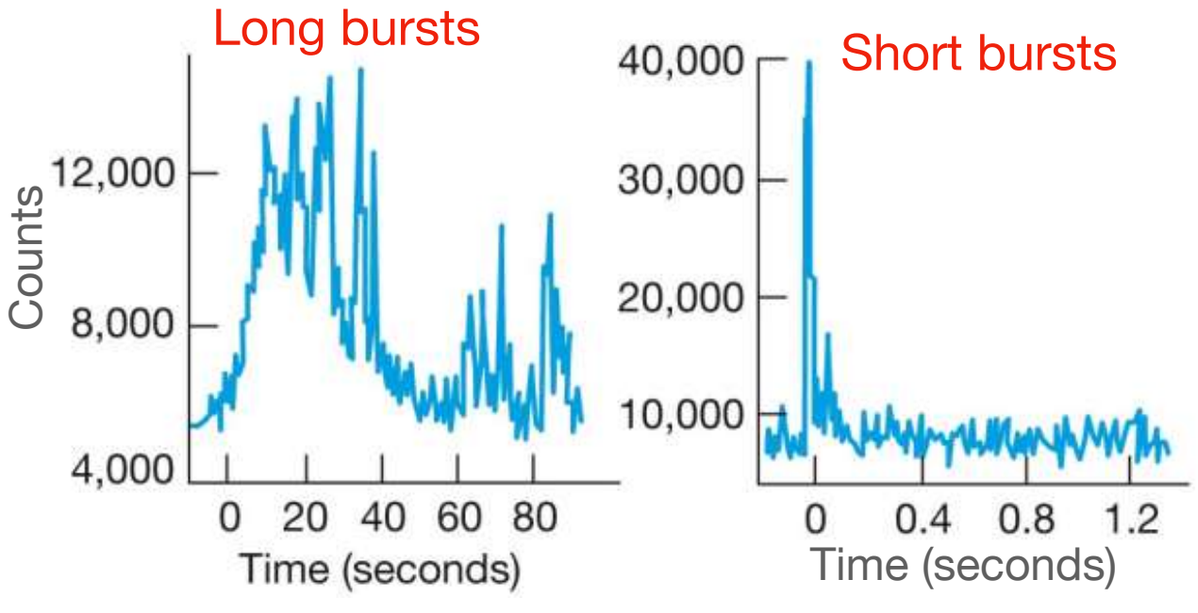
https://en.wikiversity.org/

Extragalactic objects

Gamma Ray Burst (GRB)

Gamma-ray bursts also occur, and were first spotted by satellites looking for violations of nuclear test-ban treaties. This map of where the bursts have been observed shows no "clumping" of bursts anywhere, particularly not within the Milky Way. Therefore, the bursts must originate from outside our Galaxy.





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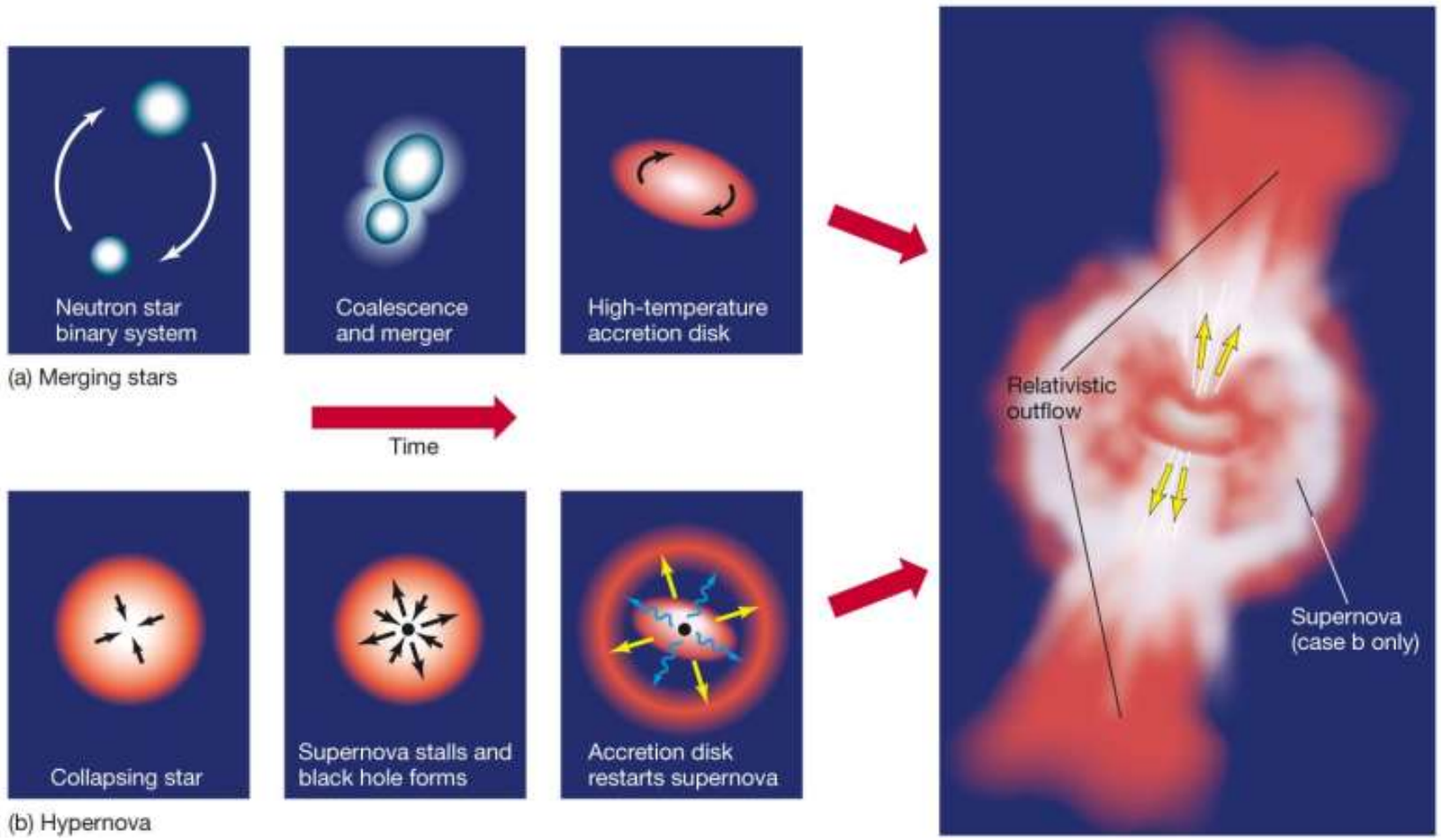
Extragalactic objects

Gamma Ray Bursts (GRBs)

Two models—merging neutron stars or a hypernova—have been proposed as the source of gamma-ray bursts:

Origin of short GRBs (Neutron stars merger)

Origin of long GRBs (Hypernova)



- GRBs are also a candidate source of ultra high-energy cosmic rays.

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Next, we will learn about the messengers in high-energy astrophysics.